Name: Kyle Salgado-Gouker

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Summary/Ethical Implications

Several challenges and ethical considerations emerged in the analysis of amusement park data that encompasses ride queue times, park attendance, and roller coaster details. The Roller Coaster Database flat file presented structural deficiencies such as missing columns, bad rows, and general formatting issues. As a result, this required meticulous data cleansing and restructuring to ensure the accuracy and coherence of the subsequent analyses.

Another significant obstacle arose from the industry's protective stance on certain data, notably a roller coaster's cost. This limitation highlighted the ethical dimensions of data access and usage, emphasizing the importance of respecting proprietary information while conducting research in sensitive sectors. Despite this, the project adapted to these constraints by focusing on what was readily available, the demand by consumers, expressed in how long they were willing to wait for a ride, a measure of popularity flux.

The API Queue Time data, a crucial component for understanding ride popularity, posed a unique challenge. Lacking a daily API, historical averages were used instead, which, as a side effect, stabilized the sparse data. The principal advantage of this data is that it is crowd-sourced and, therefore, more transparent than industry-provided data.

Ethical considerations loomed prominently throughout the project, especially in the context of data accuracy and reliability. Relying on user-reported wait times introduced potential biases, as guests might have waited for reasons unrelated to ride popularity, such as during ride breakdowns. Additionally, the inherent subjectivity in self-reported data opens the door to exaggerations, thus impacting the credibility of the analysis, but this concern is also diminished by using averages over time to measure crowd flux.

Further ethical complexities emerged concerning the influence of amusement parks on data integrity. Parks, motivated by various factors such as shareholder appeasement, could potentially manipulate numbers, leading to distorted insights. This underscores the importance of transparency and accountability in data reporting and demands researchers to critically assess the reliability of the information at hand. In other words, this is the best analysis within the constraints of the available data, and access to industry-provided data may increase obfuscation rather than diminish it.

The reliance on the Roller Coaster Database, itself relying on Wikipedia data, unveiled another layer of ethical considerations. Wikipedia is crowd-sourced, like the Queue Times data. Its collaborative and "free to edit" nature introduces other vulnerabilities to data manipulation, emphasizing the necessity for vigilant oversight to maintain the integrity of the information.

In conclusion, this amusement park analytics project showcases the iterative nature of data analysis, requiring adaptability and resourcefulness in the face of structural and access challenges. The ethical dimensions surrounding data accuracy, reliability, and industry constraints underscore the need for a nuanced approach to ensure the integrity of research outcomes and insights. The results, especially in regards to the imminent Six Flags/Cedar Fair merger, speak for themselves.